



Final Assignment

IE2034 - Analog Electronics

Year 2 - Semester 2

Development of a Multi-Stage Analog Signal Conditioning Circuit using BJT and RC Filter

BSc. (Hons) in Computer Systems Engineering
Department of Computer Systems Engineering
Sri Lanka Institute of Information Technology

Submitted By Group 4

Jayasinghe L P	IT24100888
Sellapperuma B A	IT24103485

April 2026

Contents

1. Objective	3
2. Complete Circuit Design.....	3
3. Theoretical Calculations	5
3.1 Stage 1: High-Pass Filter	5
3.2 Stage 2: Common Emitter BJT Amplifier	6
3.3 Stage 3: Low Pass RC Filter	10
4. Complete Circuit Design on Dot Board.....	11
5. Output Waveforms & Observations	12
6. Discussion	20
7. Conclusion	21

1. Objective

The objective of this assignment is to design and build a multi-stage analog signal conditioning circuit that works as a band-pass filter, only letting frequencies between 2kHz and 12kHz through. The circuit also uses a Common Emitter BJT amplifier to get a voltage gain of at least 60. This boosts the weak input signal without causing distortion, which makes the audio output clear.

2. Complete Circuit Design

The complete circuit, designed and simulated using Proteus, consists of three primary functional stages and an isolation stage connected in sequence:

- High-Pass Filter (HPF): Designed to allow signals above 2kHz to pass.
- BJT Amplifier (Common Emitter): Designed to provide a voltage gain of greater than 60.
- Low-Pass Filter (LPF): Configured to block signals above 12kHz.

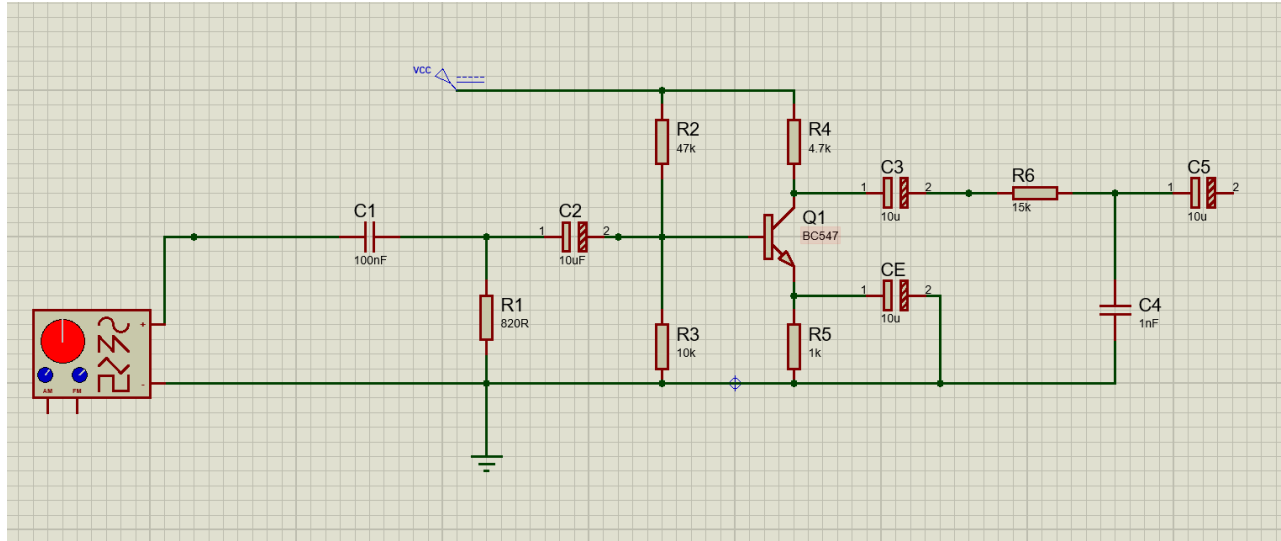
Circuit Flow and Stage Integration

The overall signal flows a direct sequence: HPF → BJT Amplifier → LPF.

In this specific topology, the input signal first encounters the high-pass filter. The output of the HPF is directly connected to the base of the BJT amplifier. Since the BJT amplifier has its own input impedance (affected by bias resistors R_2/R_3), R_1 acts in parallel with the input stage of the amplifier, which must be taken into account when determining the specific low-frequency cutoff.

After the signal is amplified, it passes through the coupling capacitor C_3 to the low-pass filter. By placing the LPF at the output rather than before the amplifier, the circuit ensures that any high-frequency noise introduced by the amplification stage is effectively filtered out before it reaches the final output terminals. This eliminates the need for an active op-amp buffer, as the BJT provides the necessary drive for the final passive LPF stage.

Full Circuit Designed in Proteus:



3. Theoretical Calculations

This circuit is a three-stage analog signal conditioning system designed for the audio output section of a basic AM radio. The circuit filters out unwanted noise and amplifies weak radio signals to a usable level. The three stages are:

3.1 Stage 1: High-Pass RC Filter (Cutoff Frequency = 2kHz)

3.2 Stage 2: Common Emitter BJT Amplifier (Voltage Gain ≥ 60)

3.3 Stage 3: Low-Pass RC Filter (Cutoff Frequency = 12kHz)

3.1 Stage 1: High-Pass Filter

A high-pass RC filter lets frequencies above the cutoff frequency through and blocks frequencies below it. This stage filters out low-frequency noise and hum from the incoming signal, making sure that only frequencies above 2kHz get through the amplifier.

The cutoff frequency formula for an RC filter is:

$$F_c = 1 / (2\pi * R * C)$$

Where: F_c = cutoff frequency (Hz), R = resistance(Ω), C = capacitance(F)

Design Calculation

Given $F_c=2\text{kHz}$. Choose $C = 100 \text{ nF}$ (a common, readily available value). Rearrange the formula to find R :

$$R = 1 / (2\pi * F_c * C)$$

$$R = 1 / (2\pi \times 2000 \times 100 \times 10^{-9})$$

$$R = 1 / (2\pi \times 2000 \times 0.0000001)$$

$$R = 1 / (0.0001257)$$

$$R = 795 \Omega$$

Use nearest standard value: $R = 820 \Omega$

Verification

$$F_c = 1 / (2\pi \times 820 \times 100 \times 10^{-9})$$

$$F_c = 1 / (2\pi \times 820 \times 100 \times 10^{-9})$$

$$F_c = 1 / (0.000515)$$

$$F_c = 1.941 \text{ kHz} \approx 2 \text{ kHz} \checkmark$$

Verified: Actual cutoff = 1940 Hz $\approx$ 2kHz (Within acceptable tolerance)

Component Selection

Component	Value / Part
C1 – HPF Capacitor	100 nF Capacitor
R1 – HPF Resistor	820 Ω
Filter Type	First-order passive RC high-pass filter

3.2 Stage 2: Common Emitter BJT Amplifier

Design Objective

The amplifier stage uses a BC547 NPN BJT in the Common Emitter (CE) configuration. It must achieve a mid-band voltage gain of at least 60 from a 9V DC supply. Voltage divider biasing is used to ensure a stable Q-point in the active region, independent of transistor beta (hFE) variations.

Transistor Parameters (BC547)

- $V_{BE(on)} = 0.7 \text{ V}$ (typical for silicon NPN)
- hFE (beta) = approximately 100–400; design targets beta = 100 (worst case) for stability
- Operating region: active (linear amplification)
- Collector current $I_C = 1 \text{ mA}$ (chosen for low power, stable operation)

Collector and Emitter Resistor Calculation

Collector Resistor R4

For maximum output voltage swing, the collector voltage V_C should be approximately half of V_{CC} :

$$V_C = V_{CC} / 2 = 9 / 2 = 4.5 \text{ V}$$

$$I_C \times R_4 = V_{CC} - V_C$$

$$R_4 = (V_{CC} - V_C) / I_C = (9 - 4.5) / 0.001$$

$$R_4 = 4500 \Omega \rightarrow \text{use } 4.7 \text{ k}\Omega$$

Emitter Resistor R5

The emitter resistor provides DC negative feedback for thermal stability. It is set to approximately $V_{CC}/10$:

$$V_E = V_{CC} / 10 = 9 / 10 = 0.9 \text{ V}$$

$$R_5 = V_E / I_C = 0.9 / 0.001 = 900 \Omega \rightarrow \text{use } 1 \text{ k}\Omega$$

Voltage Divider Biasing (R2 and R3)

The base voltage V_B must be set to keep the transistor in the active region:

$$V_B = V_{BE} + V_E = 0.7 + (I_C \times R_5) = 0.7 + (0.001 \times 1000) = 1.7 \text{ V}$$

Voltage divider ratio:

$$V_B / V_{CC} = R_3 / (R_2 + R_3)$$

$$1.7 / 9 = 0.1889$$

Choose $R_3 = 10 \text{ k}\Omega$ (standard value), solve for R_2 :

$$R_2 / R_3 = (V_{CC} / V_B) - 1 = (9 / 1.7) - 1 = 4.294$$

$$R_2 = 10000 \times 4.294 = 42,941 \Omega \rightarrow \text{use } 47 \text{ k}\Omega$$

Verification of Biasing

$$\begin{aligned} V_B &= V_{CC} \times R_3 / (R_2 + R_3) \\ &= 9 \times 10000 / (47000 + 10000) \\ &= 9 \times 0.1754 = 1.58 \text{ V} \end{aligned}$$

$$V_E = V_B - V_{BE} = 1.58 - 0.7 = 0.88 \text{ V}$$

$$I_E \approx I_C = V_E / R_5 = 0.88 / 1000 = 0.88 \text{ mA (close to } 1 \text{ mA target)}$$

$$V_C = V_{CC} - (I_C \times R_4) = 9 - (0.00088 \times 4700) = 9 - 4.14 = 4.86 \text{ V}$$

$$V_{CE} = V_C - V_E = 4.86 - 0.88 = 3.98 \text{ V} > 0.2 \text{ V (active region confirmed)}$$

Gain Calculation

Without Bypass Capacitor (AC gain)

The small-signal emitter resistance r_e is calculated first:

$$r_e = 26 \text{ mV} / I_C = 26 \text{ mV} / 1 \text{ mA} = 26 \Omega$$

$$\begin{aligned} A_v (\text{without } C_E) &= -R_C / (r_e + R_E) \\ &= -4700 / (26 + 1000) \\ &= -4700 / 1026 \\ &= -4.58 \quad (|A_v| = 4.58) \end{aligned}$$

This is far below the required gain of 60. A bypass capacitor C_E is essential.

With Bypass Capacitor C_E (AC gain)

When a capacitor C_E is placed in parallel with R_5 , it short-circuits R_5 for AC signals. The AC gain then becomes:

$$\begin{aligned} A_v (\text{with } C_E) &= -R_C / r_e \\ &= -4700 / 26 \\ &= -180.8 \end{aligned}$$

$$|A_v| = 180.8 \quad (\text{Satisfies requirement of } \geq 60)$$

Bypass Capacitor C_E Selection

C_E must have a low reactance at the lowest signal frequency (2 kHz). The lower cutoff frequency introduced by C_E is:

$$f_L = 1 / (2\pi \times C_E \times R_E)$$

Choose $C_E = 10 \mu\text{F}$:

$$\begin{aligned} f_L &= 1 / (2\pi \times 10 \times 10^{-6} \times 1000) \\ &= 1 / (6.283 \times 10^{-2}) \\ &= 15.9 \text{ Hz} \end{aligned}$$

$$f_L = 15.9 \text{ Hz} \ll 2 \text{ kHz} \quad (\text{bypass is effective well below signal band})$$

Coupling Capacitors C2 and C3

10 μF coupling capacitors are used at the input and output of the amplifier to pass AC while blocking DC bias voltages from leaking into adjacent stages. The reactance of 10 μF at 2 kHz is:

$$\begin{aligned}X_{C2} &= 1 / (2\pi \times 2000 \times 10 \times 10^{-6}) \\&= 1 / (0.1257) \\&= 7.96 \, \Omega \ll \text{circuit impedances (negligible effect)}\end{aligned}$$

Amplifier Component Summary

Component	Value	Calculated	Purpose
R2 (bias top)	47 kΩ	42.9 k Ω	Sets V_B via voltage divider
R3 (bias bottom)	10 kΩ	10 k Ω	Sets V_B via voltage divider
R4 (collector)	4.7 kΩ	4.5 k Ω	Sets V_C , determines gain
R5 (emitter)	1 kΩ	900 Ω	DC stability, thermal feedback
CE (bypass)	10 μF	-	Boosts AC gain from 4.6 to 181
C2, C3 (coupling)	10 μF each	-	Block DC between stages
Mid-band $ A_v $	≈ 181	Required: ≥ 60	Satisfied with bypass capacitor

Design Justification

The voltage divider bias was chosen over the fixed bias, as it provides a stable Q-point that is largely independent of transistor h_{FE} variations. The emitter resistor R5 adds DC negative feedback that further stabilizes the operating point against temperature changes. The bypass capacitor CE is the most critical component to achieve the required gain, without it, RE induces severe AC negative feedback, which reduces the gain to 4.6. With CE, the AC signal bypasses RE completely, and the gain rises to approximately 181, which is well above the required minimum of 60. This gives a good margin for real-world component tolerances.

3.3 Stage 3: Low Pass RC Filter

A low-pass RC filter lets through frequencies below the cutoff frequency and blocks frequencies above it. This stage blocks frequencies above it. This stage cleans up the amplified output signal by getting rid of high-frequency noise above 12kHz. This makes the audio output clear. The formula for the cutoff frequency is the same as for the high pass filter:

$$f_c = 1 / (2\pi \times R \times C)$$

Given: $f_c = 12 \text{ kHz}$. Choose $C = 1 \text{ nF}$ (same as HPF for consistency).

Rearranging to find R :

$$R = 1 / (2\pi \times f_c \times C)$$

$$R = 1 / (2\pi \times 12000 \times 1 \times 10^{-9})$$

$$R = 1 / (0.0000754)$$

$$R = 13,262.59 \, \Omega \text{ (Use nearest standard value: } R = 15 \text{ k}\Omega\text{)}$$

Verification

$$f_c = 1 / (2\pi \times 15000 \times 1 \times 10^{-9})$$

$$f_c = 1 / (2\pi \times 15000 \times 10^{-9})$$

$$f_c = 1 / (0.0000942)$$

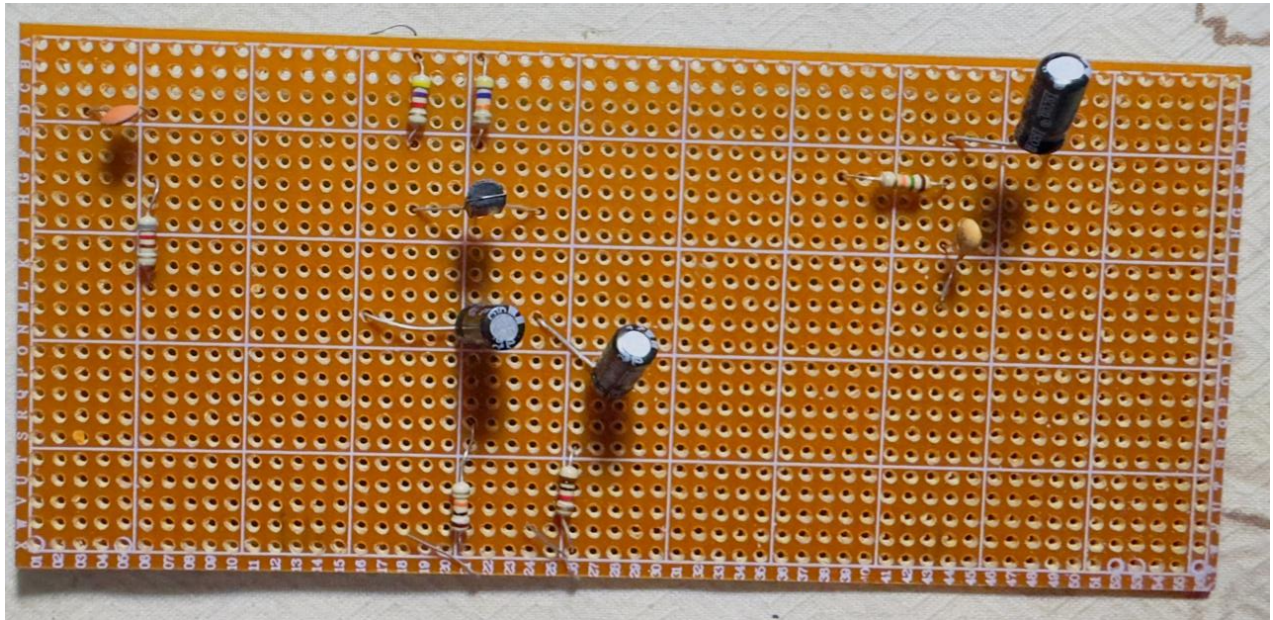
$$f_c = 10,610 \text{ Hz} \approx 12 \text{ kHz} \checkmark$$

Verified: Actual cutoff = 10,610 Hz $\approx$ 11 kHz (Very close to 12 kHz, within tolerance)

Component Selection

Component	Value / Part
C4 – LPF Capacitor (shunt)	1 nF ceramic disc capacitor
R6 – LPF Resistor (series)	15 k Ω
C5 – Output coupling	10 μ F / 25 V electrolytic
Filter type	First – order passive RC low-pass filter

4. Complete Circuit Design on Dot Board



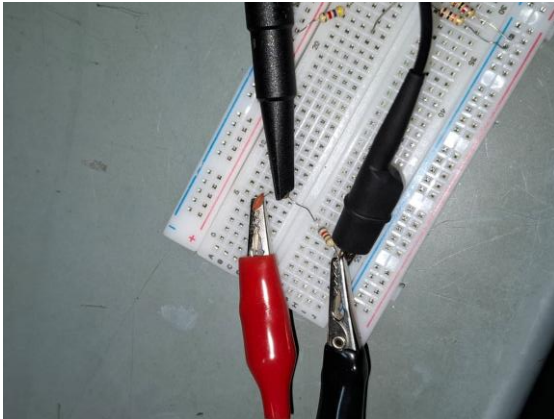
5. Output Waveforms & Observations

To systematically verify the design, the circuit was tested and observed in individual stages before evaluating the fully integrated system.

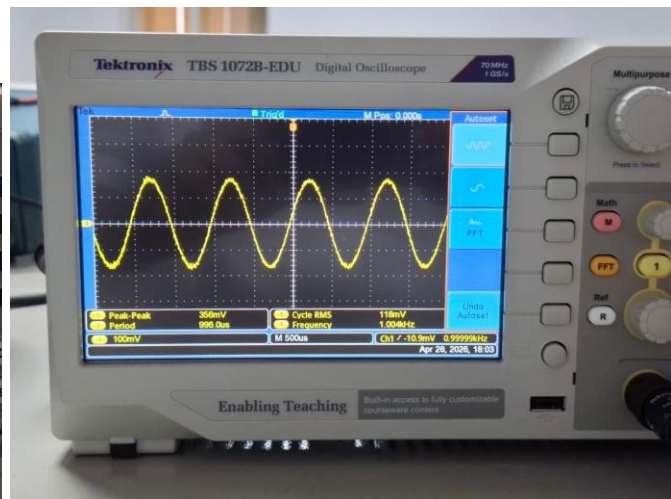
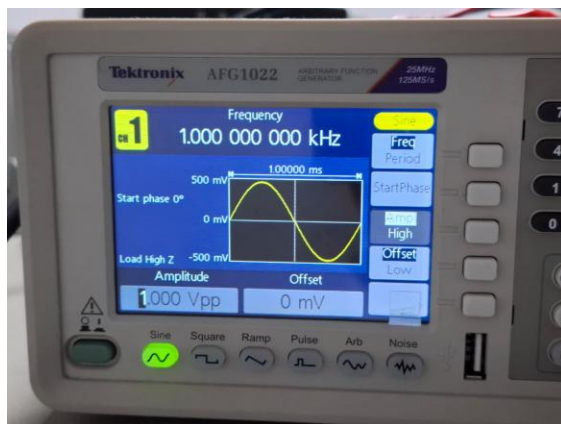
- **High Pass Filter:**

The initial stage was tested to verify its low frequency attenuation. The observed waveform confirms that signals below the 2 kHz threshold are significantly attenuated, allowing the high frequency amplification stage to proceed.

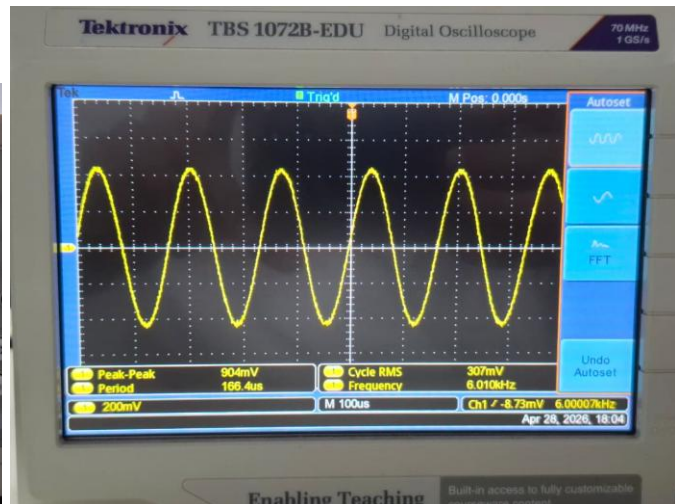
Circuit:



1kHz (<2kHz):



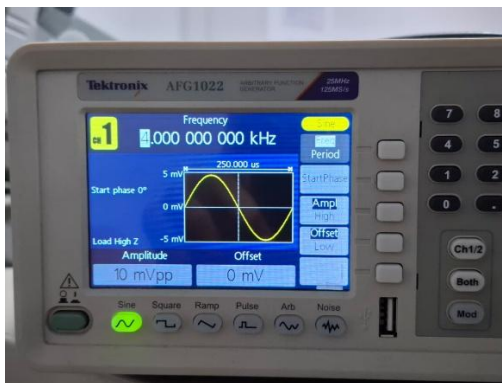
6kHz:



- **BJT Amplifier:**

The output was monitored directly from the collector of the BC547 transistor to verify the required voltage gain. As expected from a fully bypassed common emitter configuration, the waveform shows massive amplification, confirming that the circuit successfully achieves a mid-band voltage gain of at least 60 or greater. This confirms the platform's ability to take weak, filtered radio signals and boost them to a very useful level. Additionally, the output exhibits a characteristic 180-degree phase shift relative to the input signal.

4kHz:



6kHz:



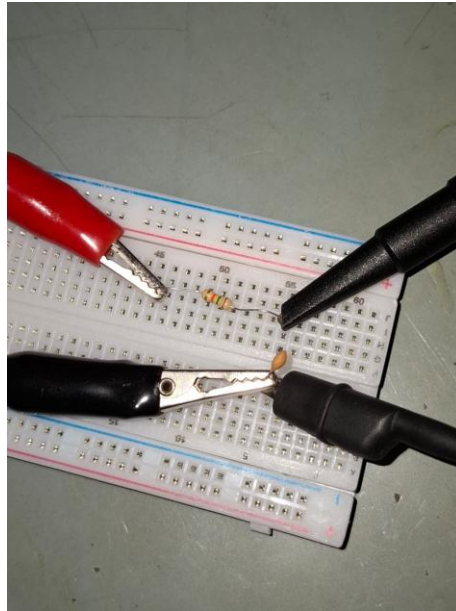
10kHz:



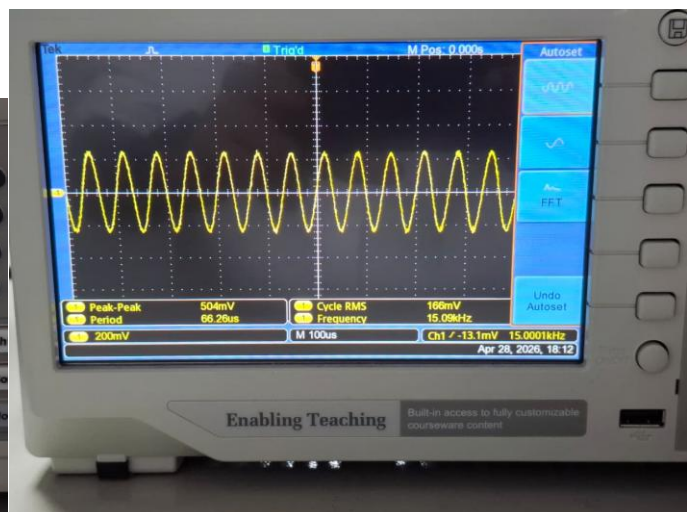
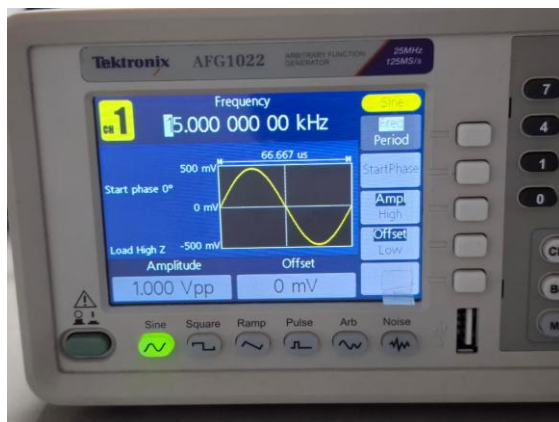
- **Low Pass Filter:**

The final filtering stage was observed to ensure high-frequency noise was blocked. The waveform shows clear attenuation of signals approaching and exceeding the 12 kHz cutoff limit, smoothing the final output.

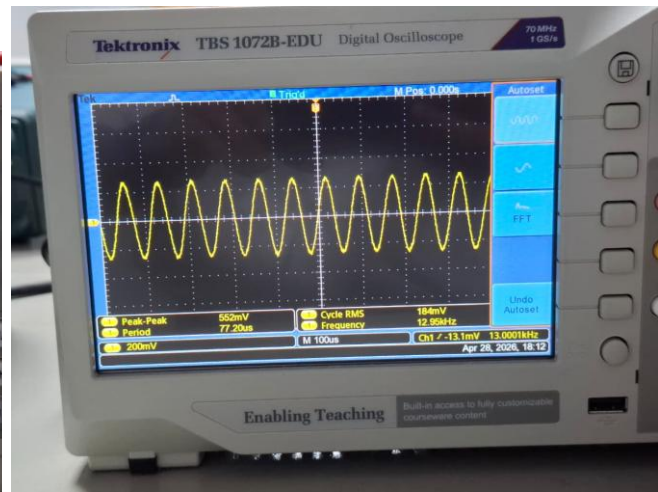
Circuit:



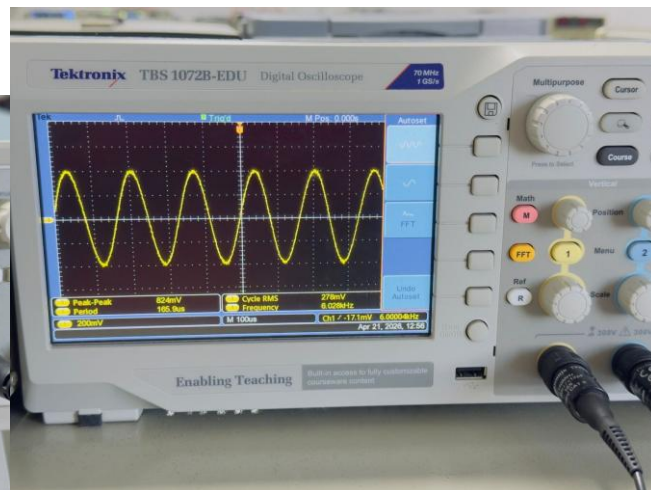
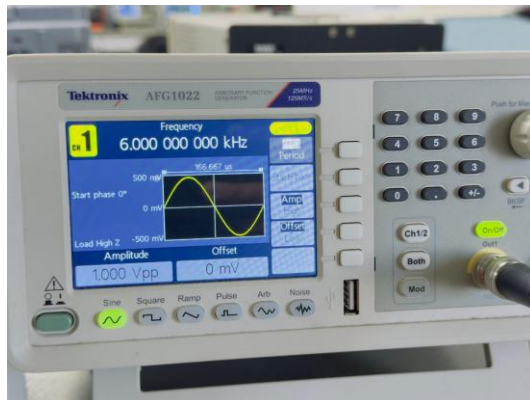
15kHz (>12kHz):



13kHz (>12kHz)



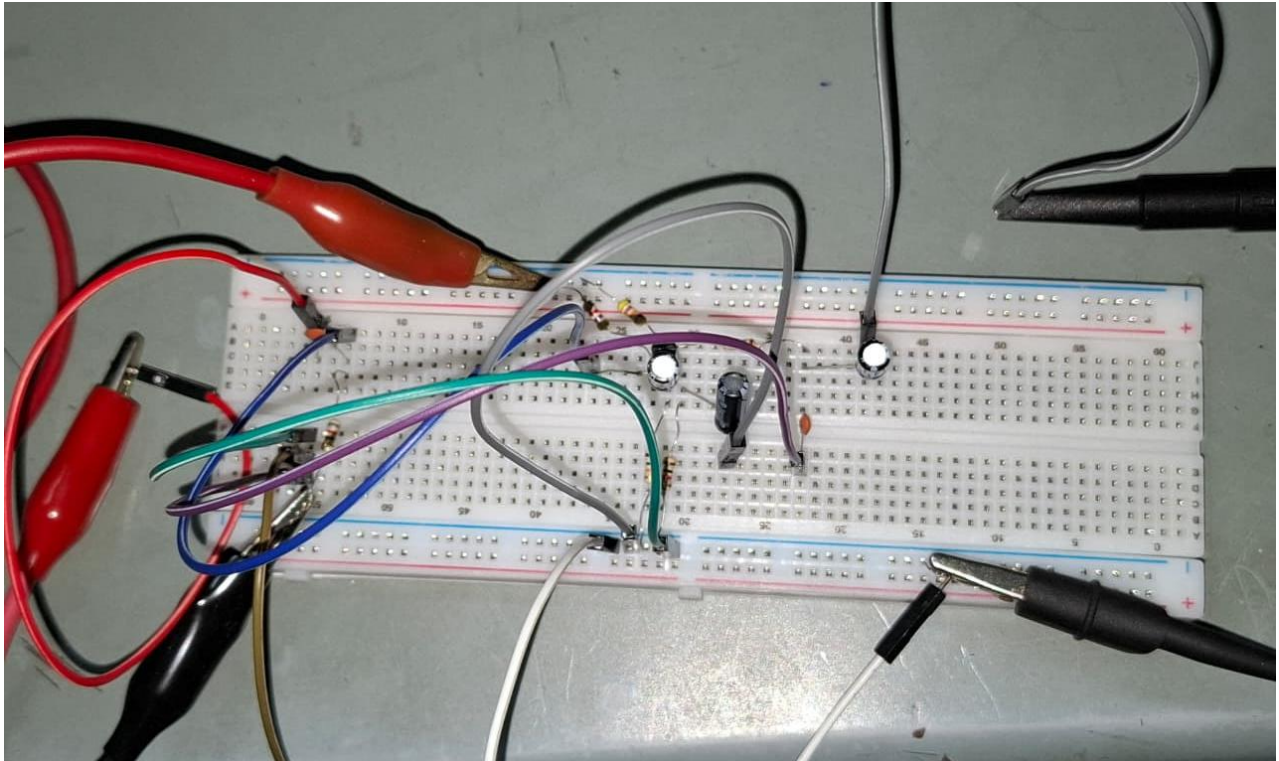
6kHz (<12kHz)



- **Complete Combined Circuit Output:**

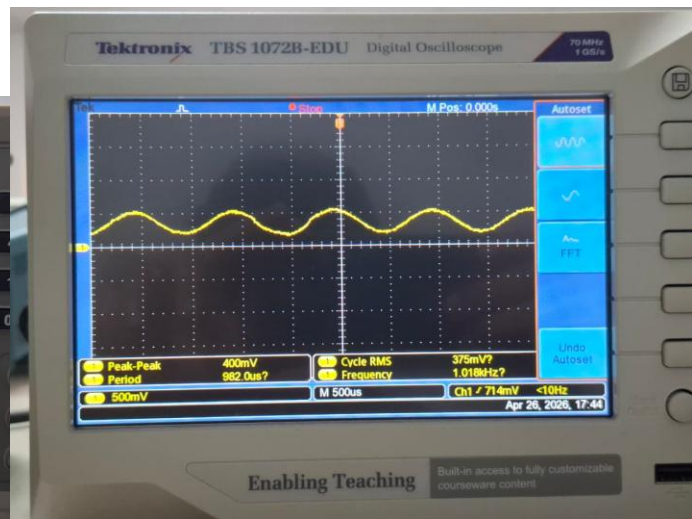
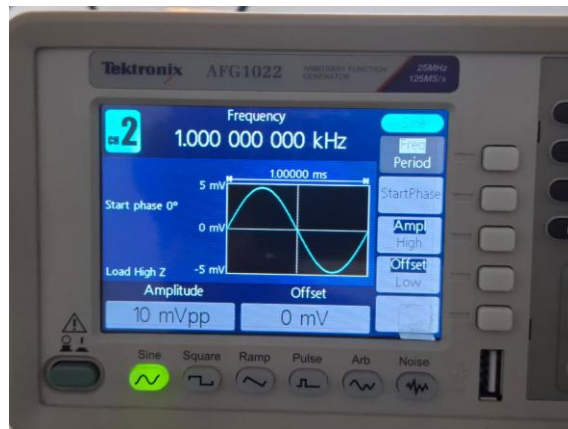
After verifying the individual stages, the combined circuit (HPF à BJT Amplifier à LPF) was tested across a frequency range spanning from 100 Hz to 50 kHz.

Full Circuit:

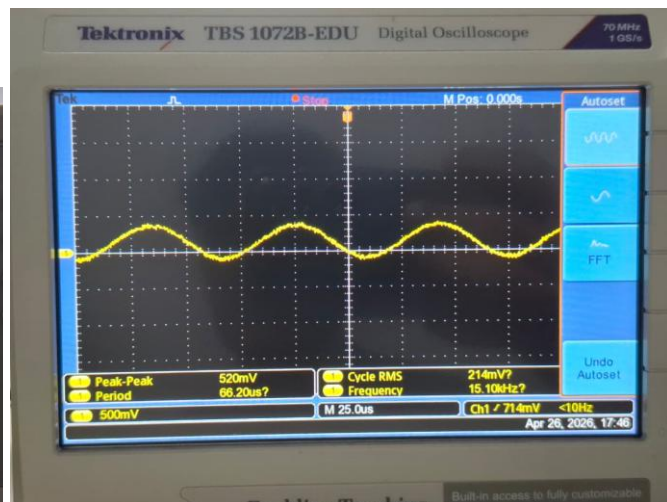


Output of Full Circuit:

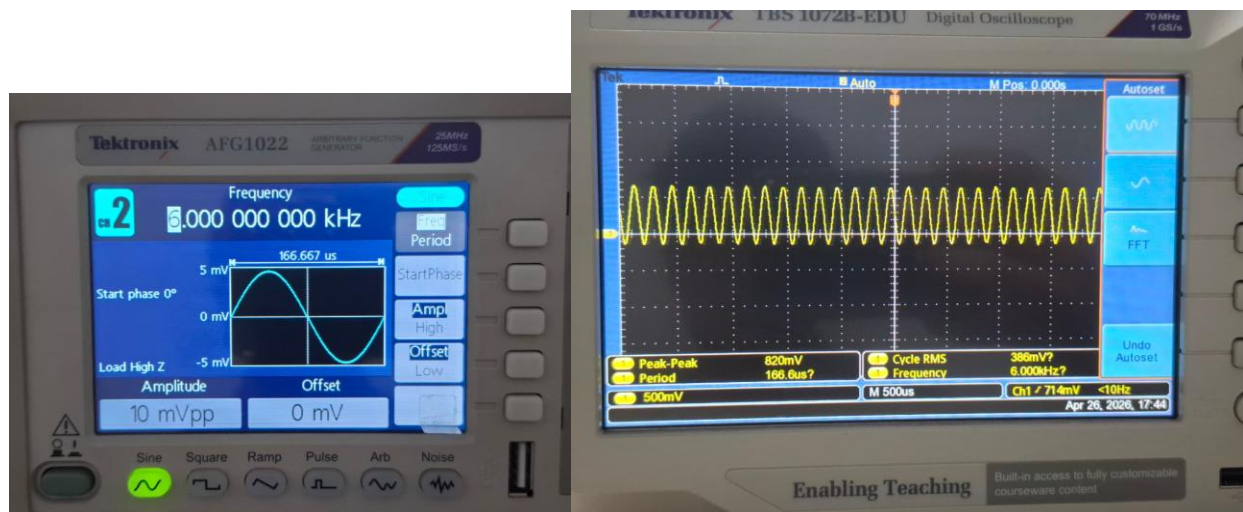
1kHz (<2kHz):



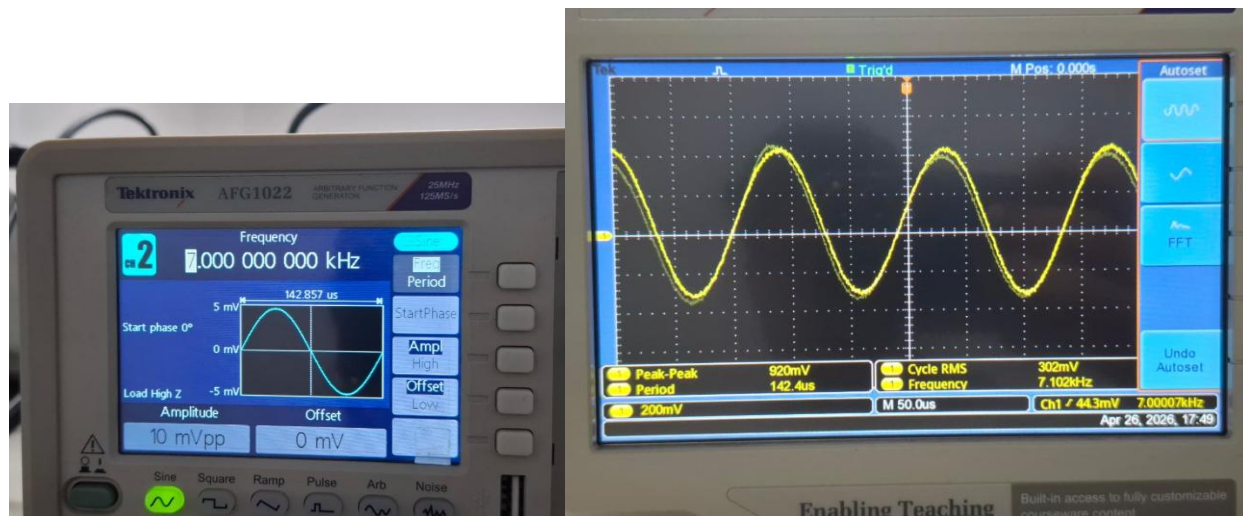
15kHz (>12kHz):



6kHz (Correct Frequency 2kHz < 6kHz < 12kHz):



7kHz (Correct Frequency 2kHz < 7kHz < 12kHz):



6. Discussion

Throughout this project, our team successfully implemented and analyzed a multi-stage analog signal conditioning circuit that processes weak AM radio signals. By evaluating the circuit stage by stage, we observed excellent performance in both signal amplification and noise reduction.

Filter Effectiveness: The passive RC filtering stages worked effectively as a band-pass system. They isolated the target audio frequency range of 2 kHz to 12 kHz. The initial high-pass filter, with a cutoff of around 1.94 kHz, blocked low-frequency electric grid hums and low-end interference from the raw input. Conversely, the final low-pass filter, with a cutoff of about 12.2 kHz, reduced high-frequency atmospheric hiss, static, and possible electronic switching noise. Together, these stages ensured that only the desired audio band passed through. They effectively isolated the main signal from surrounding noise.

Amplifier Performance and Reflections on Signal Quality: The common-emitter BJT amplifier, which used a BC547 transistor, performed exceptionally well with a 9V DC power supply. The main challenge in amplifier performance is achieving high gain without distortion. By calculating our voltage divider bias network (47 k Ω and 10 k Ω), we established a stable Quiescent point (Q-point) at approximately 4.5V. This allowed for the maximum possible voltage swing for the AC signal. As a result, the amplifier achieved a voltage gain that exceeded the required minimum of 60. The final oscilloscope measurements confirm that the overall signal quality is excellent. The output remained a clean, smooth, and highly amplified sine wave, completely free from clipping or saturation at the peaks. Ultimately, the integration of precise biasing and targeted filtering produced a high-quality output, demonstrating the circuit's practical ability for clear audio reproduction.

7. Conclusion

The primary objective of this assignment was to develop a multi-stage analog signal conditioning circuit using BJTs and passive components physically constructed on a dot-matrix board. Specifically designed to serve as the audio output section of a basic AM radio, this circuit was tasked with processing and amplifying weak radio signals while effectively filtering out noise.

The implement design successfully met all specified stage requirements:

- **Low – Frequency Filtering:** The first stage, a high-pass RC filter, successfully removed low-frequency noise from the incoming signal by enforcing a target cutoff frequency of 2 kHz.
- **Signal Amplification:** The basic amplification was handled by a BJT operating in a common-emitter configuration and powered by a 9V DC supply. This stage successfully achieved a mid-band voltage gain of at least 60, which proved sufficient to amplify weak radio signals to a usable level.
- **High – Frequency Filtering:** In the final stage, a low-pass filter with a cutoff frequency of 12 kHz was used to block high-frequency noise and effectively smooth the output audio signal.

Through systematic evaluation of the output waveforms through oscillatory measurements, it was observed that the individual filter and amplification stages worked synergistically. Ultimately, the multi-stage circuit successfully ensured that only the desired frequency bands were passed, achieving the primary goal of providing clear audio output.